

Power Electronics

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Desk D216

Layout

- Non isolated choppers
- **Switch mode power supplies**
 - Introduction
 - Flyback
 - Forward
 - Push-Pull
- Power components



Introduction

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Introduction



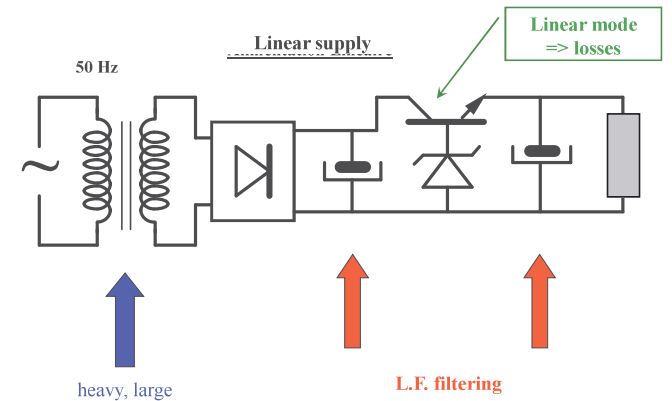
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Linear isolated power supply principle



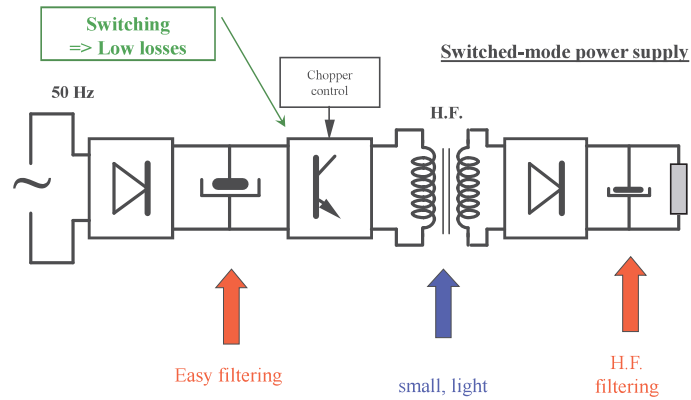
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Isolated switched-mode power supply principle



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Comparison

Element	LINEAR	Switching
Transformer	Large (50 Hz)	Small (H.F.)
Transistor	Linear (losses)	Switching
Output filter	Large (50 Hz)	Simple (H.F.)
Command	autonomous, simple	control, complex
Efficiency	40 % < < 50 %	70 % < < 90 %
Control	0,1 %	0,5 %

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Switched-mode power supply nomenclature

- **Flyback** : energy storage converter
- **Forward** : direct converter
- **Push-Pull** : symmetric converter (full bridge or half bridge)

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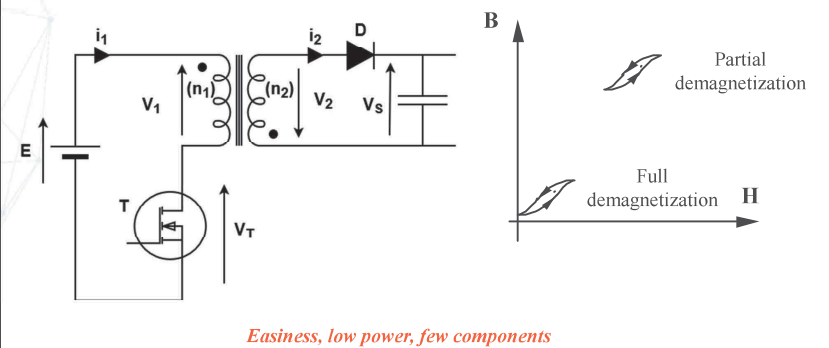
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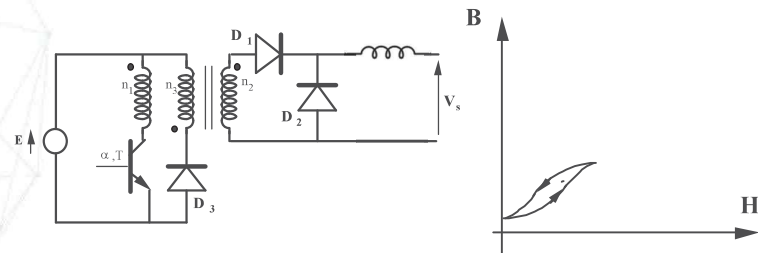
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Forward principle



Magnetic component optimized, more components

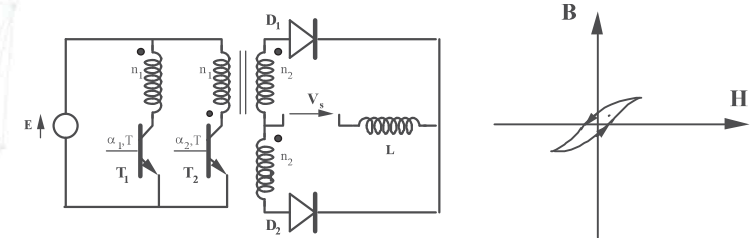
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Symmetric converter



High power, 2 transistors or more

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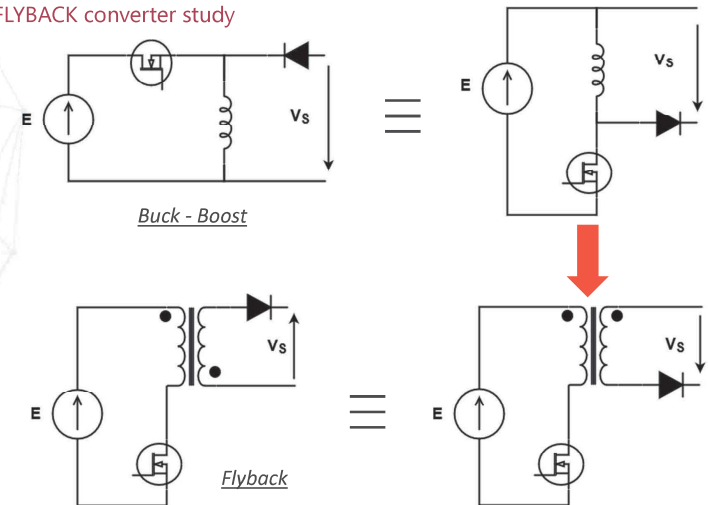
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FLYBACK converter study



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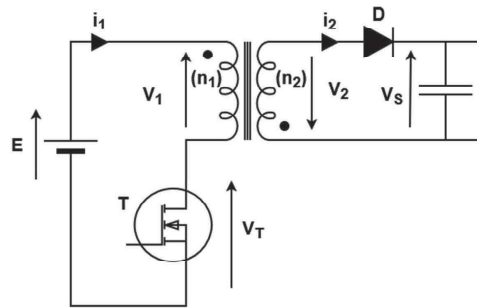
Sizing

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FLYBACK converter study



Hypothesis : E constant, V_s constant, perfect semi-conductors

2 magnetic modes

Partial demagnetization (continuous mode)

Full demagnetization (discontinuous mode)

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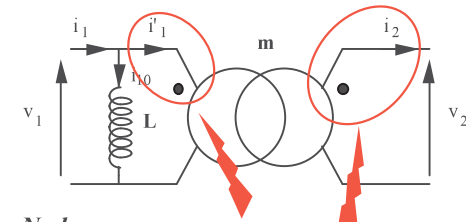
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Transformer model: convention !



No losses

L : magnetizing inductance (at the primary)

R : Reluctance of the magnetic circuit

$$m = \frac{n_2}{n_1} = \frac{v_2}{v_1} = \frac{i_1'}{i_2}$$

$$\oint H \cdot dl = R \cdot \varphi = n_1 \cdot i_1 - n_2 \cdot i_2 = n_1 \cdot i_{10}$$

Hopkinson's law

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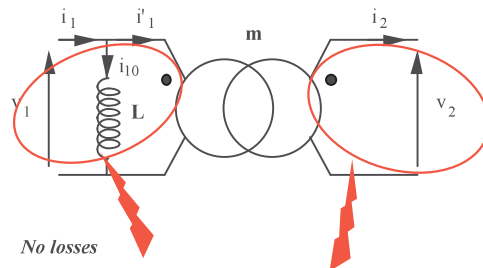
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Transformer model: convention !



No losses

$$v_1 = +n_1 \frac{d\varphi}{dt} \quad v_2 = +n_2 \frac{d\varphi}{dt}$$

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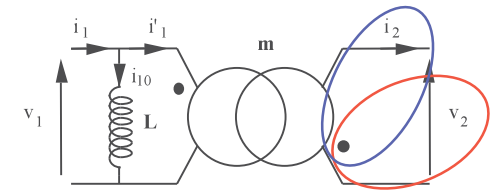
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FLYBACK transformer model



$$m = \frac{N_2}{N_1} = -\frac{v_2}{v_1} = -\frac{i_1'}{i_2}$$

$$v_2 = -n_2 \frac{d\varphi}{dt}$$

$$\oint H \cdot dl = R \cdot \varphi = n_1 \cdot i_1 + n_2 \cdot i_2 = n_1 \cdot i_{10}$$

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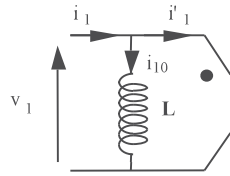
Sizing

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Push-Pull



Magnetizing current



$$i_1 = i_{10} + i'_1$$

$$R \cdot \phi = n_1 \cdot i_{10} \quad L_1 = \frac{n_1^2}{R}$$

Magnetizing current $\Leftrightarrow$ flux in the magnetic circuit

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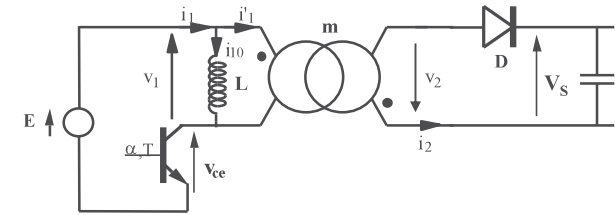
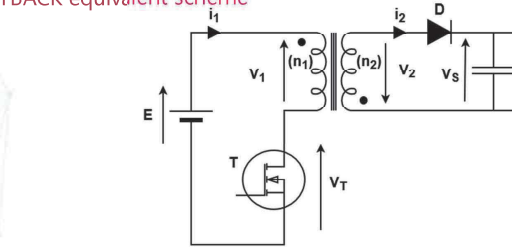
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FLYBACK equivalent scheme



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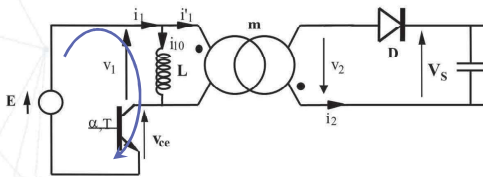
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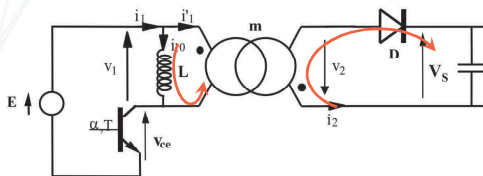
Push-Pull



Partial demagnetization mode: continuous conduction



$0 < t < \alpha T : T_p \text{ closed}$
 $\Rightarrow v_1 = E \Rightarrow v_2 = m \cdot E$
 $v_D = -mE - V_S < 0 \Rightarrow D \text{ reverse}$
 $i_2 = 0 \Rightarrow i'_1 = 0 \Rightarrow i_1 = i_{10}$
 Energy storing



$\alpha T < t < T : T_p \text{ opened}$
 $\Rightarrow i_1 = 0 \Rightarrow i'_1 = -i_{10} < 0$
 $\Rightarrow i_2 < 0 \Rightarrow D \text{ forward}$

$v_2 = -V_S \Rightarrow v_1 = -V_S/m \Rightarrow \phi \text{ decreases}$
 Energy restitution

$$\oint H \cdot dl = R \cdot \phi = n_1 \cdot i_1 - n_2 \cdot i_2 = n_1 \cdot i_{10}$$

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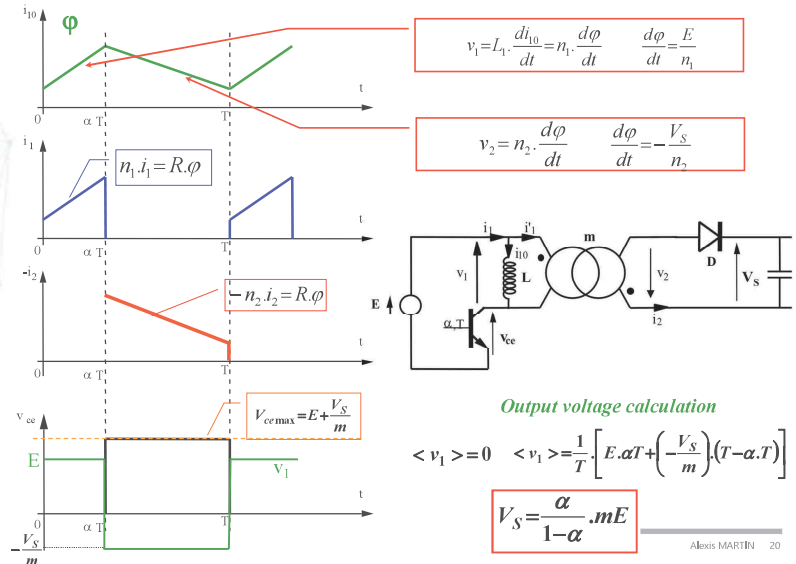
Continuous mode

Discontinuous mode

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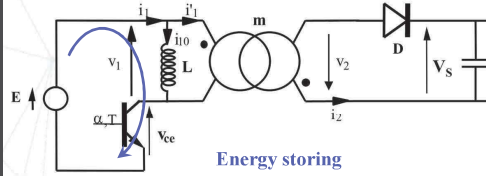
Sizing

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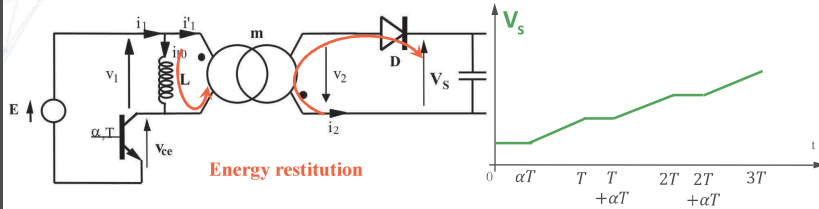
Push-Pull



Demagnetization issue



Energy storing



Energy restitution

Warning:

If no load connected
-> Output capacitor never discharges
-> Issue

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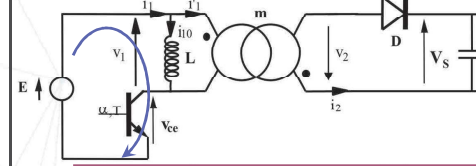
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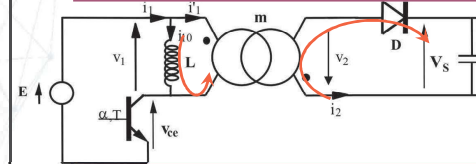


Discontinuous mode: behavior



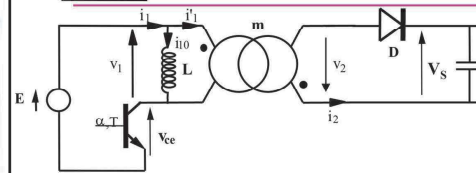
$0 < t < \alpha T : T_p \text{ closed}$
 $\Rightarrow v_1 = E \Rightarrow v_2 = m.E$
 $v_D < 0 \Rightarrow D \text{ reverse}$

$i_2 = 0 \Rightarrow i'_1 = 0 \Rightarrow i_1 = i_{10}$
Energy storage
Unchanged mode



$\alpha T < t < \alpha' T : T_p \text{ opened}$
 $\Rightarrow i_1 = 0 \Rightarrow i'_1 = -i_{10} < 0$
 $\Rightarrow i_2 < 0 \Rightarrow D \text{ forward}$
and then reverse naturally

$v_2 = -V_S \Rightarrow v_1 = -V_S/m \Rightarrow \phi \text{ decreases}$
Energy restitution



$\alpha' T < t < T : T \text{ and } D \text{ both reverse}$
All currents are zero
No more energy
 $\Rightarrow v_1 = 0 ; v_2 = 0 \Rightarrow \phi = 0$

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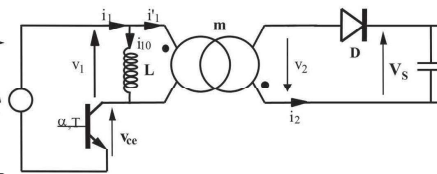
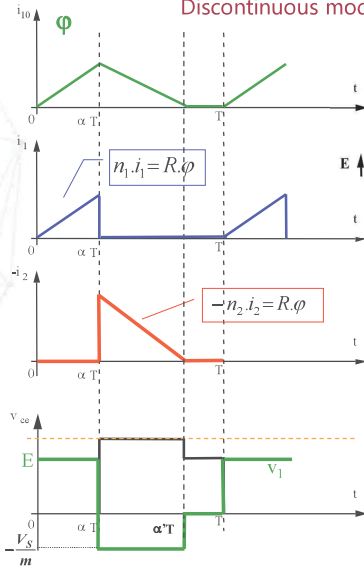
Sizing

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Discontinuous mode: chronograms



Output voltage calculation:
Transferred energy

$$W_L = \frac{1}{2} \cdot L \cdot I_{Lmax}^2 = \frac{1}{2} \cdot L \cdot \left(\frac{\alpha \cdot E \cdot T}{L} \right)^2$$

$$P_S = V_S \cdot I_S = f \cdot W_L$$

$$V_S = \frac{\alpha^2 \cdot E^2 \cdot T}{2 \cdot L \cdot I_S}$$

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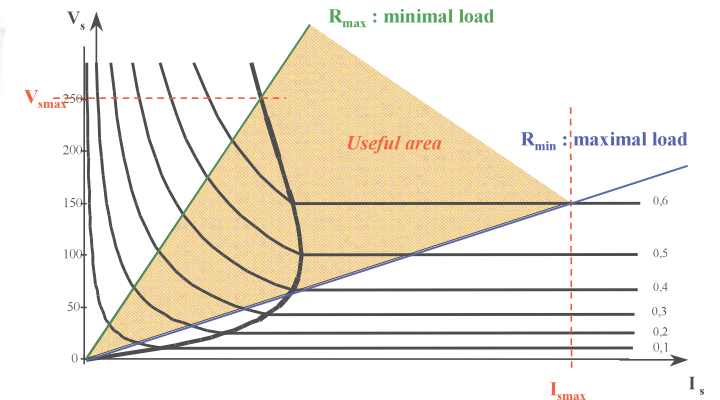
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Discontinuous mode: output characteristics



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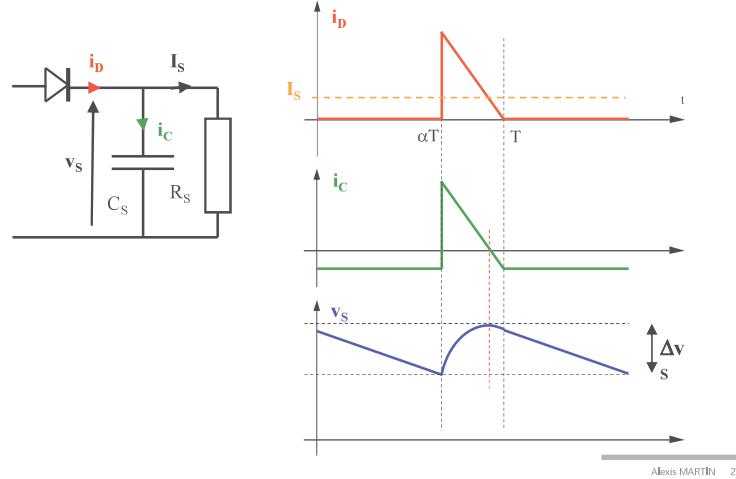
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Boundary mode: Output filtering



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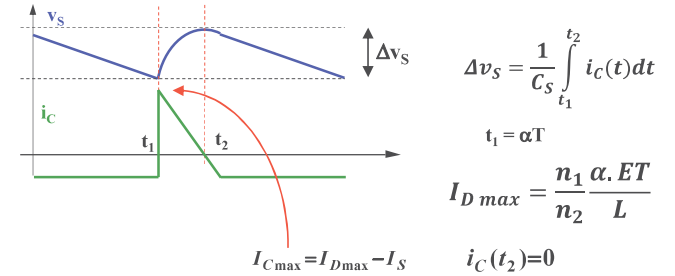
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Boundary mode: capacitor calculation



$$C_S = \frac{\alpha^2 \cdot E \cdot T}{(1 - \alpha) \cdot R_S \cdot \Delta V_S}$$

same as
Buck-Boost

For high value of C_S ,
 $\Delta v_S \approx \Delta v_{ESR} = ESR \cdot \Delta i_C = ESR \cdot \Delta i_L$

$$ESR < \frac{\Delta V_S}{\Delta i_L}$$

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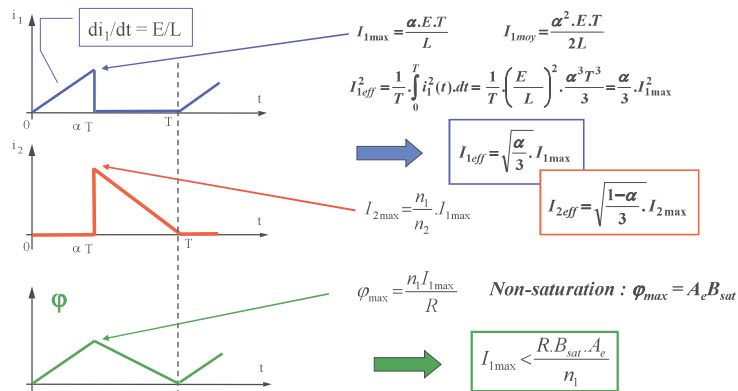
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Boundary mode: sizing components



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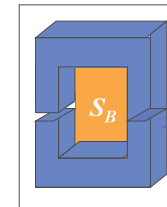
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Boundary mode: transformer sizing



$$\delta = \frac{I_{1eff}}{s_{sp1}} = \frac{I_{2eff}}{s_{sp2}} < \text{max current density: } \delta_{max}$$

$$S_{Tcu} = n_1 \cdot s_{sp1} + n_2 \cdot s_{sp2} < k_B \cdot \text{Window area } S_B$$

k_B = winding coefficient

$$k_B \cdot S_B > n_1 \cdot \frac{I_{1eff}}{\delta_{max}} + n_2 \cdot \frac{I_{2eff}}{\delta_{max}}$$

$$I_{1eff} = \sqrt{\frac{\alpha}{3}} \cdot I_{1max}$$

$$I_{2eff} = \sqrt{\frac{1-\alpha}{3}} \cdot I_{2max}$$

$$I_{2max} = \frac{n_1}{n_2} \cdot I_{1max}$$

$$I_{1max} < \frac{k_B \cdot S_B \cdot \delta_{max}}{n_1} \cdot \frac{\sqrt{3}}{\sqrt{\alpha + \sqrt{1-\alpha}}}$$

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Boundary mode: transformer sizing

$$P_S = f \cdot W_L = f \cdot \frac{1}{2} \cdot L \cdot I_{1\max}^2$$

$$I_{1\max} < \frac{R \cdot B_{sat} \cdot A_e}{n_1}$$

$$I_{1\max} < \frac{k_B \cdot S_B \cdot \delta_{\max}}{n_1} \cdot \frac{\sqrt{3}}{\sqrt{\alpha + \sqrt{1 - \alpha}}}$$

$$P_S < f \cdot \frac{L}{2} \cdot \frac{R \cdot B_{sat} \cdot A_e}{n_1} \cdot \frac{k_B \cdot S_B \cdot \delta_{\max}}{n_1} \cdot \frac{\sqrt{3}}{\sqrt{\alpha + \sqrt{1 - \alpha}}}$$

$$A_e \cdot S_B > 2 \cdot \frac{\sqrt{\alpha + \sqrt{1 - \alpha}}}{\sqrt{3}} \cdot \frac{P_S}{f \cdot k_B \cdot B_{sat} \cdot \delta_{\max}}$$

With P_S
With f

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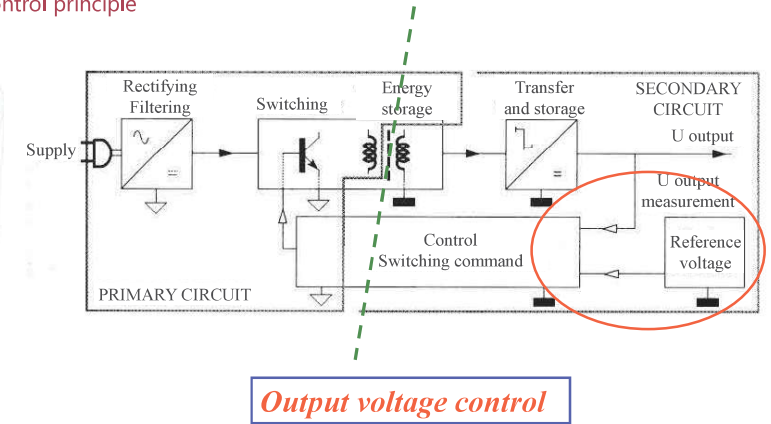
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Control principle



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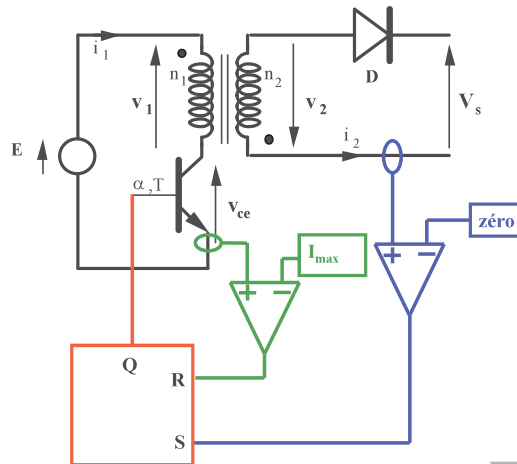
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Boundary mode holding



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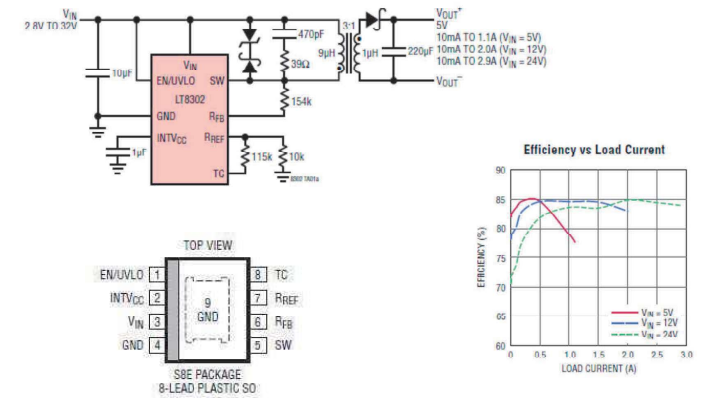
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Example: LT8302



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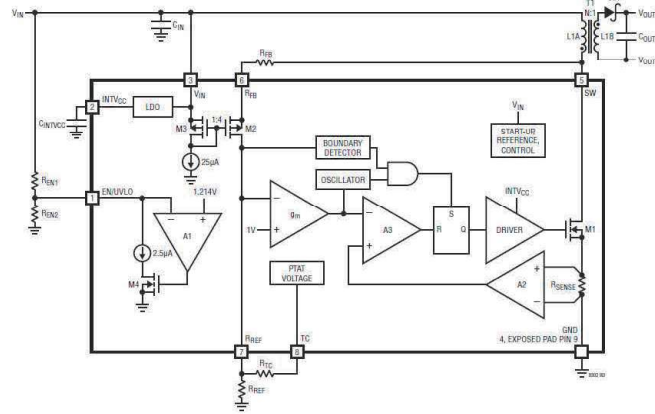
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Example: LT8302 (internal scheme)



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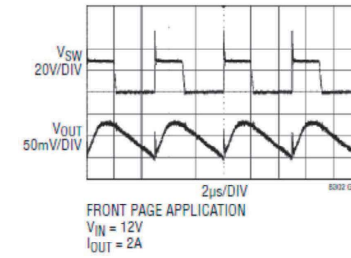
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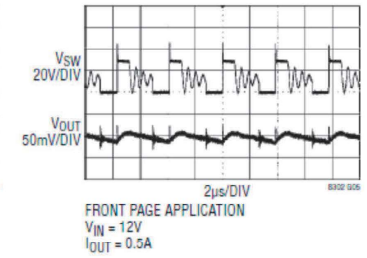


Example: LT8302

Boundary Mode Waveforms



Discontinuous Mode Waveforms



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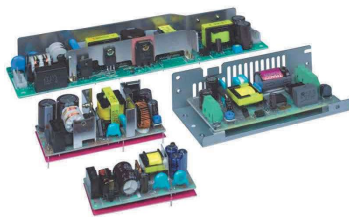
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Flyback examples



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Forward

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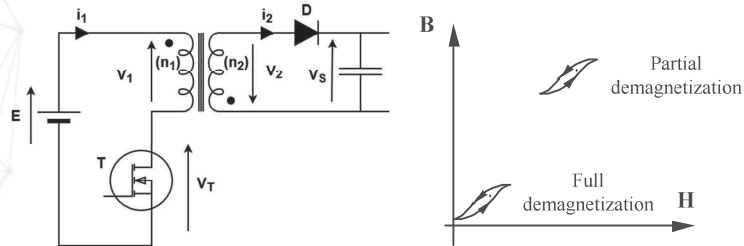
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Flyback principle



Easiness, low power, few components

Limited for high power due to energy storage limitations

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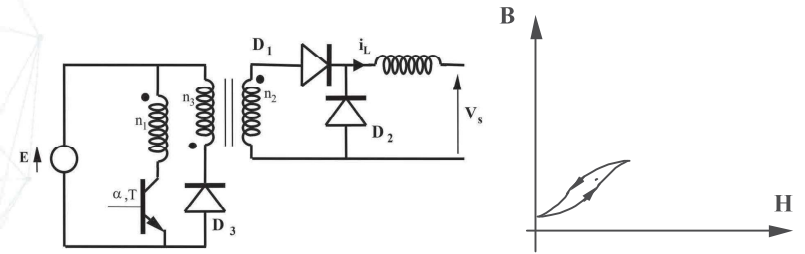
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Forward principle



Magnetic component optimized, more components

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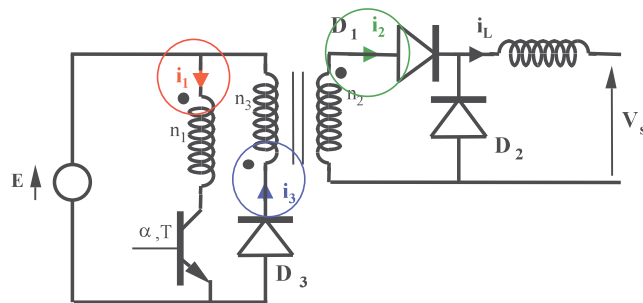
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Hopkinson's law



$$\oint H \cdot dl = R \cdot \phi = n_1 \cdot i_1 + n_3 \cdot i_3 - n_2 \cdot i_2$$

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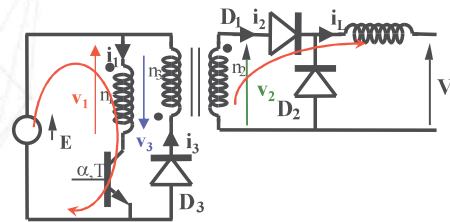
Continuous mode

Sizing

Push-Pull

Continuous mode study: $i_L > 0$

➡ At least one of the two diodes D_1 and D_2 forwards.



$$v_1 = E = n_1 \cdot \frac{d\phi}{dt} \quad \text{magnetization}$$

$$v_2 = n_2 \cdot \frac{d\phi}{dt} \quad v_3 = n_3 \cdot \frac{d\phi}{dt}$$

$0 < t < \alpha T$

T_p forward.

$$v_1 = E \Rightarrow v_2 = \frac{n_2}{n_1} \cdot E$$

➡ D_1 forward, D_2 reverse
 $i_2 = i_L > 0$

$$v_1 = E \Rightarrow v_3 = \frac{n_3}{n_1} \cdot E$$

➡ D_3 reverse

$$n_1 \cdot i_1 - n_2 \cdot i_2 = R \cdot \phi$$

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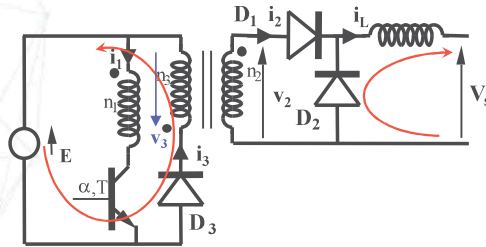
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Continuous mode study



$$T_p \text{ opened} \Rightarrow i_1 = 0$$

$$n_3 \cdot i_3 - n_2 \cdot i_2 = R \cdot \phi > 0$$

$$\Rightarrow i_3 > 0 \Rightarrow D_3 \text{ forward}$$

$$v_3 = -E = n_3 \cdot \frac{d\phi}{dt} \quad \text{demagnetization}$$

$$v_2 = -\frac{n_2}{n_3} E \quad D_1 \text{ reverse, } D_2 \text{ forward} \Rightarrow i_2 = 0$$

$$L \cdot \frac{di_L}{dt} = -V_S$$

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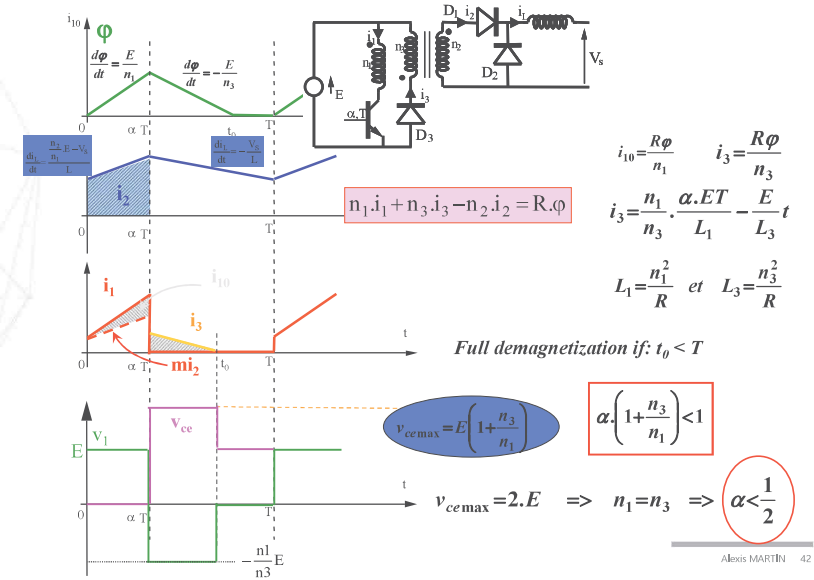
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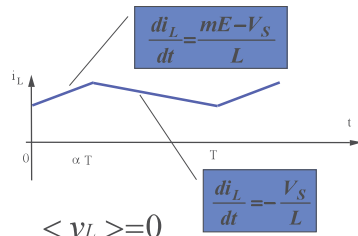
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Continuous mode study: voltage transfer



$$\Delta i_L = \frac{mE - V_S}{L} \cdot \alpha T = \frac{V_S}{L} (1 - \alpha) T$$

$$V_S = \alpha \cdot m \cdot E$$

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Continuous mode study: Components sizing

➤ **Filtering capacitor: C_S**

➤ **Transformer $\Rightarrow Ae \cdot SB$**

➤ **Output inductance $\Rightarrow Ae \cdot SB$**

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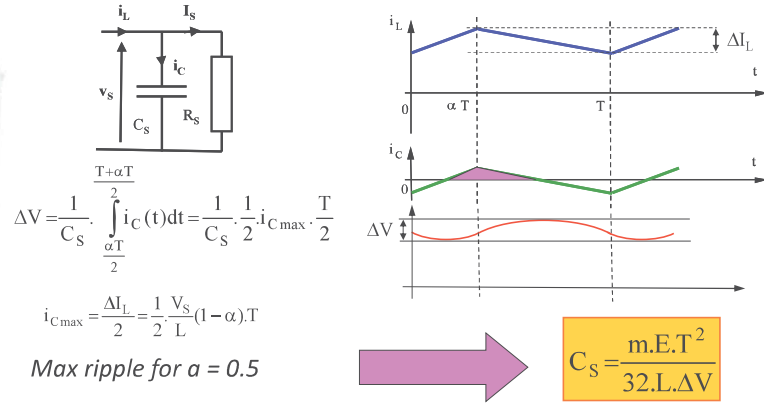
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Continuous mode study: Filtering capacitor



For high value of C_S ,
 $\Delta v_S \approx \Delta v_{ESR} = ESR \cdot \Delta i_C = ESR \cdot \Delta i_L$

$$ESR < \frac{\Delta V_S}{\Delta i_L}$$

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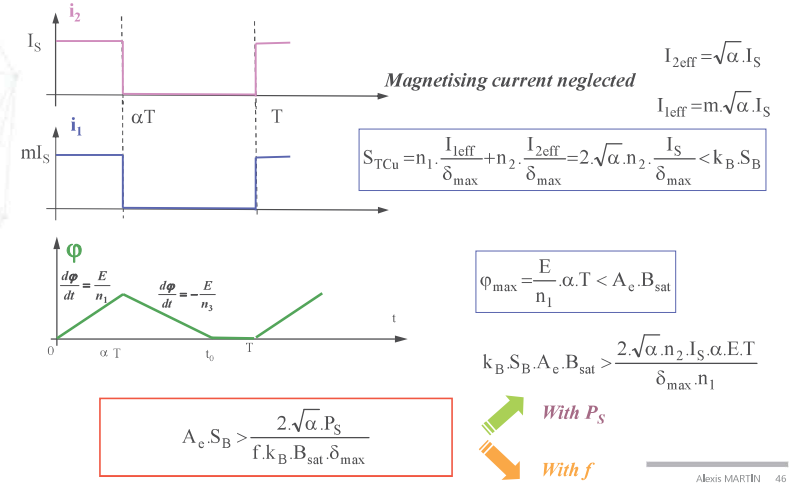
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Continuous mode study: transformer sizing



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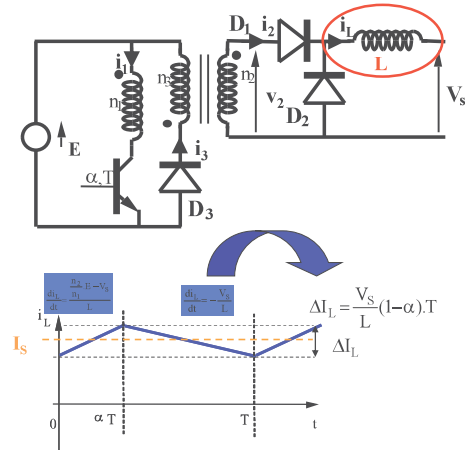
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Continuous mode study: inductor sizing



Hold the continuous mode

$$I_S > \frac{\Delta I_L}{2}$$

$$L > \frac{\alpha \cdot V_S}{2 I_S \cdot f}$$

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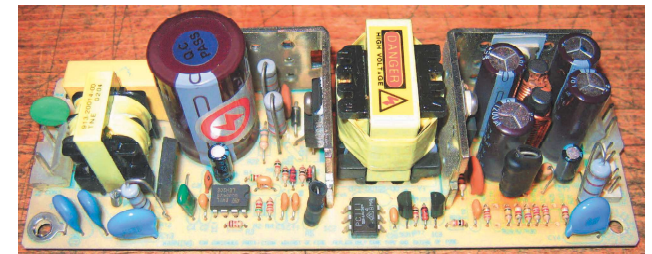
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Continuous mode
Sizing

Push-Pull



Example



Manufacturer reference: CI Design Y0-2040AA
Input : 100 à 240 Vac
Output : 12 Vdc (2 A) 5 Vdc (3 A)

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Flyback

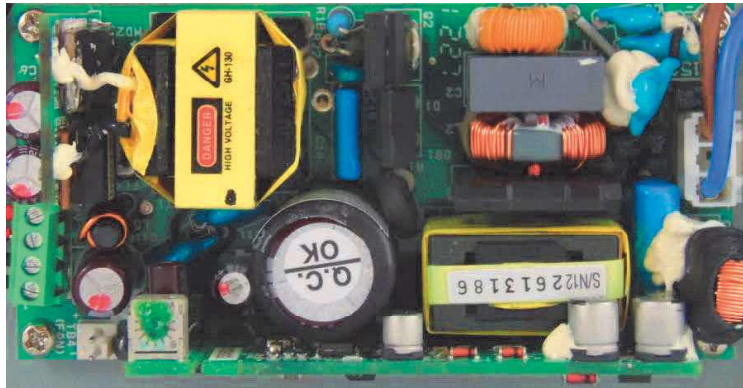
Forward

Principle
Continuous mode
Sizing

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Example



NuForce

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Push-Pull



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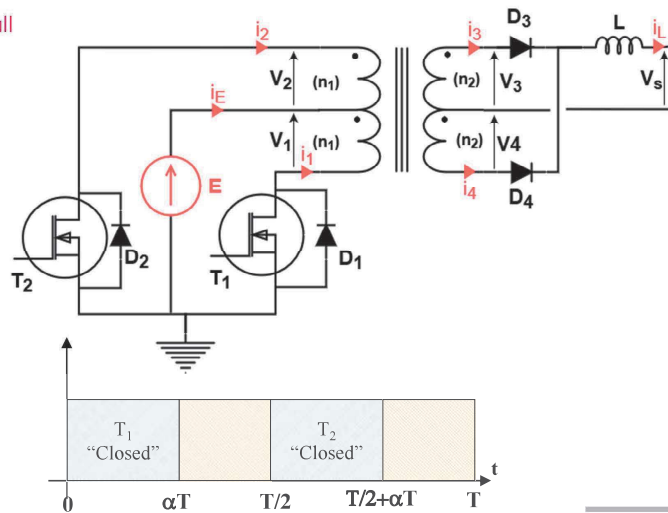
Flyback

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Push-Pull



Push-Pull



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Push-Pull

No load study
Continuous mode
Other structures

No load study : $i_L = 0$

$0 < t < \alpha T$: T_1 "closed", T_2 "open"

- $v_1 = n_1 \cdot \frac{d\phi}{dt} = E > 0$

Magnetization

- $v_2 = v_1 = E > 0$

$\Rightarrow D_2$ « reverse »

$$-n_1 \cdot i_1 = \mathcal{R} \cdot \phi$$

$\alpha T < t < T/2$: T_1 & T_2 "open"

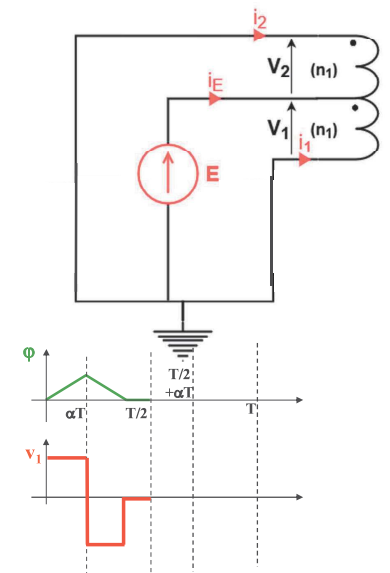
- $\mathcal{R} \cdot \phi = -n_1 \cdot i_1 + n_1 \cdot i_2 > 0$

$\Rightarrow D_2$ « forward », D_1 « reverse »

- $\mathcal{R} \cdot \phi = n_1 \cdot i_2 > 0$

- $v_2 = n_1 \cdot \frac{d\phi}{dt} = -E < 0$

Demagnetization



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No load study : $i_L = 0$

$T/2 < t < T/2 + \alpha T$: T_1 "open", T_2 "closed"

- $v_1 = n_1 \cdot \frac{d\phi}{dt} = -E < 0$

Magnetization

- $v_2 = v_1 = -E < 0$

➡ D_1 « reverse »

$$n_1 \cdot i_2 = \mathcal{R} \cdot \phi$$

$T/2 + \alpha T < t < T$: T_1 & T_2 "open"

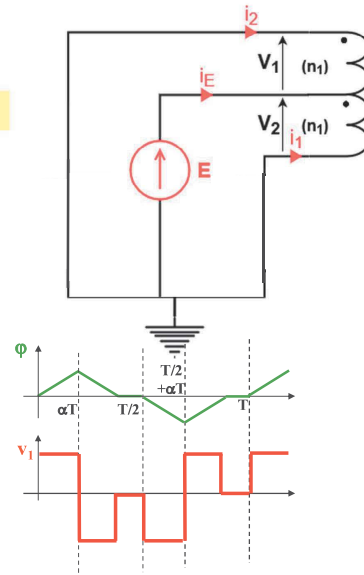
- $\mathcal{R} \cdot \phi = -n_1 \cdot i_1 + n_1 \cdot i_2 < 0$

➡ D_1 « forward », D_2 « reverse »

- $\mathcal{R} \cdot \phi = -n_1 \cdot i_1 < 0$

- $v_2 = n_1 \cdot \frac{d\phi}{dt} = E > 0$

Demagnetization



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Flyback

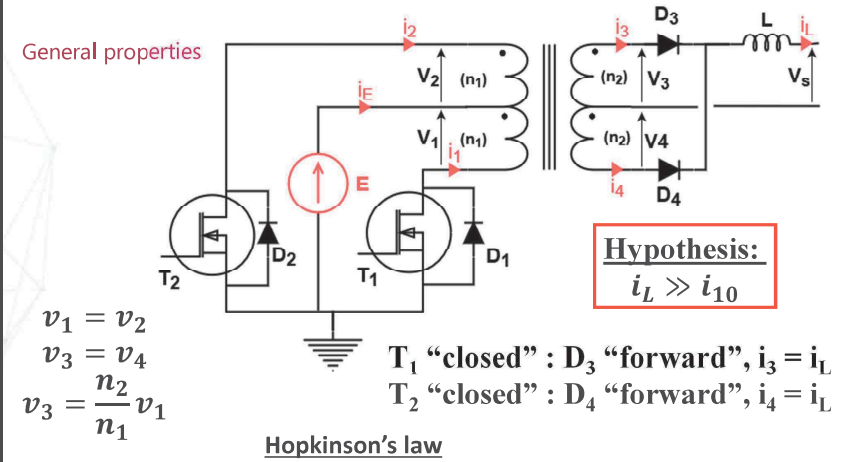
Forward

Push-Pull

No load study
Continuous mode
Other structures



General properties



$$\oint H \cdot dl = \mathcal{R} \cdot \phi = -n_1 \cdot i_1 + n_1 \cdot i_2 - n_2 \cdot i_3 + n_2 \cdot i_4$$

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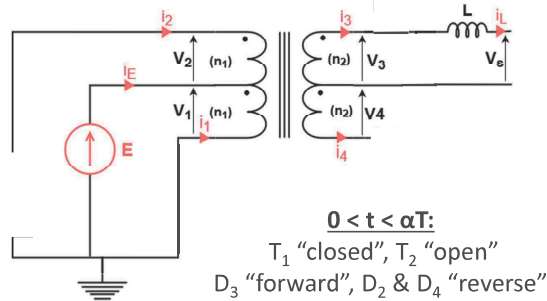
Forward

Push-Pull

No load study
Continuous mode
Other structures



Continuous mode study: magnetization

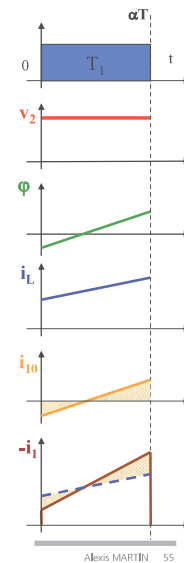


- $v_1 = n_1 \cdot \frac{d\phi}{dt} = E > 0$

Magnetization

$$-n_1 \cdot i_1 - n_2 \cdot i_3 = \mathcal{R} \cdot \phi$$

$$\Rightarrow i_1 = -\frac{\mathcal{R}}{n_1} \cdot \phi - \frac{n_2}{n_1} \cdot i_3 = -i_{10} - \frac{n_2}{n_1} \cdot i_L$$



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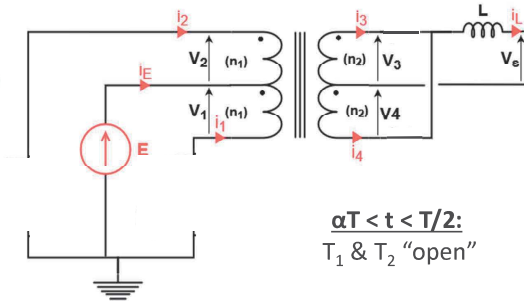
Forward

Push-Pull

No load study
Continuous mode
Other structures



Continuous mode study: second phase



$$\mathcal{R} \cdot \phi = -n_1 \cdot i_1 + n_1 \cdot i_2 - n_2 \cdot i_3 + n_2 \cdot i_4 > 0$$

- If D_2 « forward » $\rightarrow D_4$ « forward »

$$\mathcal{R} \cdot \phi = n_1 \cdot i_2 + n_2 \cdot i_L$$

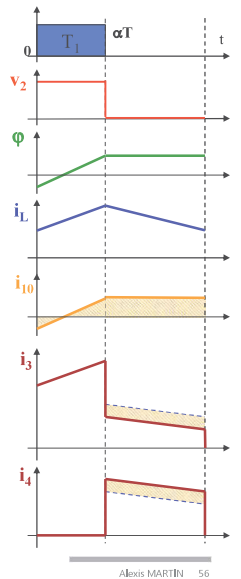
$$\mathcal{R} \cdot \phi = -n_2 \cdot i_3 + n_2 \cdot i_4 > 0$$

$$v_2 = v_3 = v_4 = 0$$

➡ D_2 "reverse"

➡ D_3 & D_4 "forward"

$$\phi = cst$$



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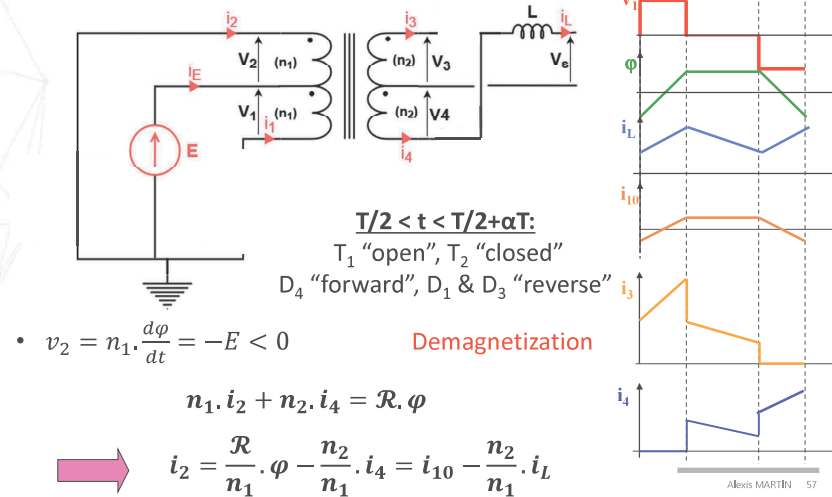
No-load study

Continuous mode

Other structures



Continuous mode study: demagnetization



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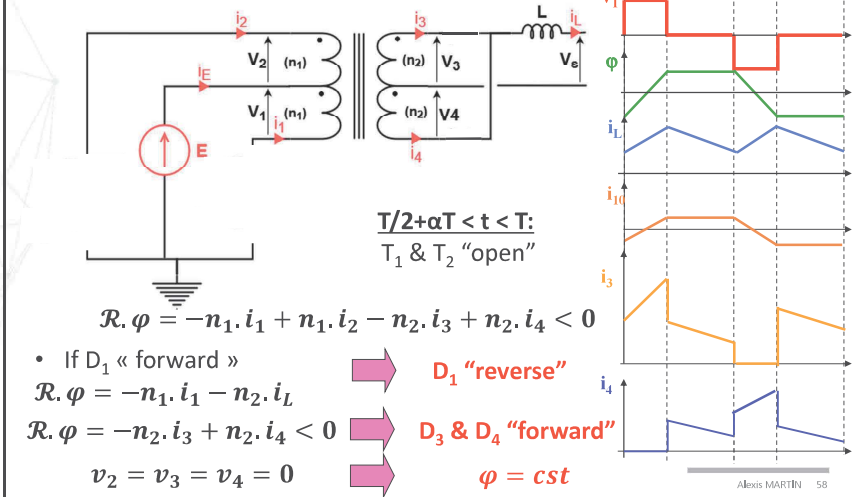
No-load study

Continuous mode

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Continuous mode study: fourth phase



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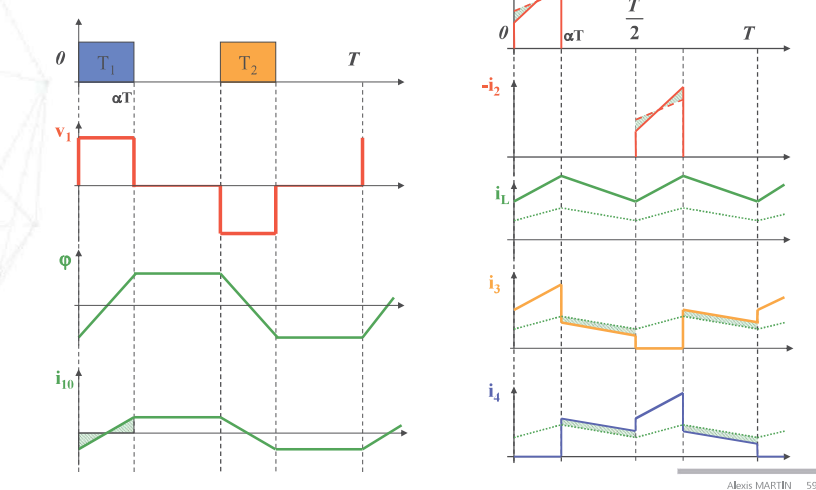
No-load study

Continuous mode

Other structures



Continuous mode study: chronograms



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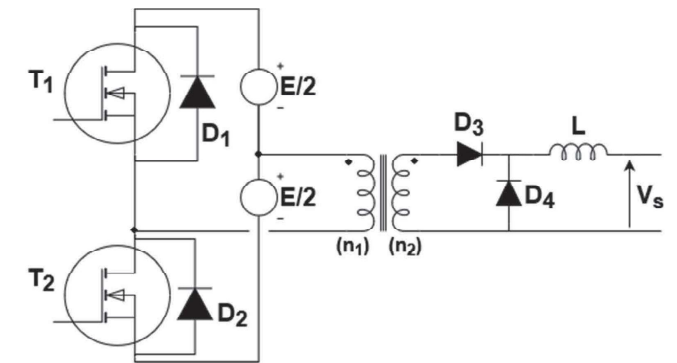
No-load study

Continuous mode

Other structures



Other symmetrical structures : Half Bridge Forward



- Same behavior as Push-Pull
- Only on primary and secondary winding
- Transformer simpler (so cheaper)

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No-load study

Continuous mode

Other structures



Other symmetrical structures: Full Bridge Forward

